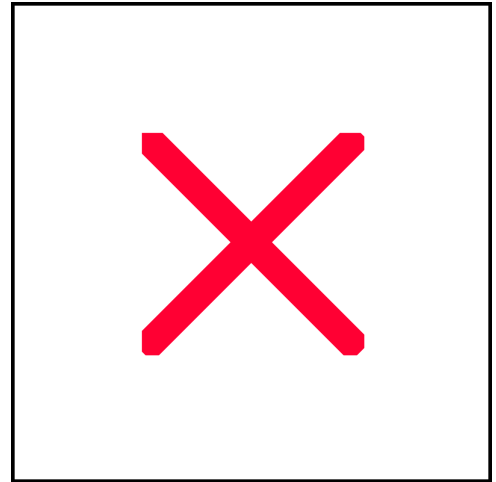
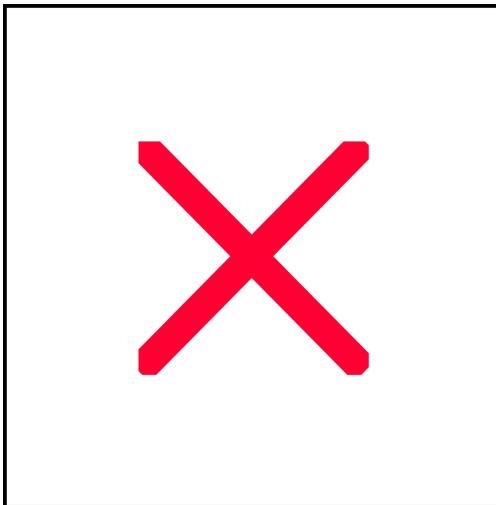




Psychology and Neuroscience 33rd Annual InHouse Conference



May 3 & 4, 2007
Life Sciences Centre
Room 4258/63

Welcome to the 2007 Psychology and Neuroscience InHouse Convention

Thanks for attending the 33rd Annual InHouse Convention.

Thanks should also be extended to the InHouse Convention Committee for planning and organization. The committee consists of the following individuals: Sophie Jacques, Nancy Bandstra, Sabrina Bellhouse, Yoko Ishigami, Tonya Kuznetsova, Tim O'Leary and Amy Wong.

Friendly Reminders

- * All talks are in LSC 4258/4263.
- * Please bring your own mug for coffee or tea!
- * All presentations are listed in the name of the presenter from the abstracts submitted.
- * There will be a reception at the end of Friday in the 2nd Floor Lounge.

Thursday, May 3, 2007

8:45-10:30 Session 1. Chair: Yoko Ishigami

- T1) Richard Brown: Introduction and brief history of convention
- T2) Sherry Stewart and Brigitte Sabourin
- T3) Ron Hoffman & Heather Schellinck
- T4) Ana Chica
- T5) Joshua Salmon

10:30-11:00 Coffee Break

11:00-12:30 Session 2. Chair: Antonina Kuznetsova

- T6) Brigitte Sabourin
- T7) Heather Fulton
- T8) Penny Corkum
- T9) Nicolle Vincent
- T10) Esther Lau
- T11) Samüel Deurveilher

12:30-1:45 Lunch Break and Posters (Psychology 2nd Floor Lounge)

- P1) Carly Prusky
- P2) Ella Laur
- P3) Jennifer Mullane
- P4) Ashley Brown

1:45-3:00 Session 3. Chair: Sabrina Bellhouse

- T12) Sarah Dreumont-Boudreau
- T13) Rachel Dingle
- T14) Sunday Bisong
- T15) Jennifer Stamp
- T16) Steve Shaw

3:00-3:30 Coffee Break

3:30-5:15 Session 4. Chair: Aimee Wong

- T17) Claudia Groh
- T18) Shini-ya Takemura
- T19) Yoshitaka Hamanaka
- T20) Tara Edwards
- T21) Joanna Borowska
- T22) Janusz Borycz
- T23) Birgit Greiner

Friday, May 4, 2007

8:45-10:30 Session 5. Chair: Antonina Kuznetsova

- T24) Marcus Juodis
- T25) Kate Thompson
- T26) Marc Zahradnik
- T27) Jonathan Fawcett
- T28) Kristin Smith
- T29) Heath Matheson
- T30) Kristie Dukewich

10:30-11:00 Coffee Break

11:00-12:30 Session 6. Chair: Yoko Ishigami

- T31) John Barresi
- T32) Chris Moore
- T33) Timothy Dummer
- T34) Cindy Hamon-Hill
- T35) Leanne ten Brinke
- T36) Deborah Johnson-Emberly

12:30-1:45 Lunch Break and Posters (Psychology 2nd Floor Lounge)

- P5) Hilary Duncan
- P6) Jaime Williams
- P7) Nicola Hoffman
- P8) Teri Bolivar

1:45-3:15 Session 7. Chair: Aimee Wong

- T37) Mora Martin
- T38) Jaime Williams
- T39) Jodie Gawryluk
- T40) Erin Mazerolle
- T41) Dominic Standage
- T42) Nathan Saruk

3:15-3:45 Coffee Break

3:45-5:15 Session 8. Chair: Sabrina Bellhouse

- T43) Jonathan Fawcett
- T44) Meredith Flannery
- T45) Yoko Ishigami
- T46) Lisa Wright
- T47) Rahia Mashoodh

T48) Richard Brown

5:30 Dinner/Reception in the 2nd Floor Lounge

TALKS

T1) Historical Introduction to the In-House Conference

Richard E. Brown

Welcome to the 33rd annual Psychology/Neuroscience In-House Conference. This presentation will introduce you to the people who started the In-House Conference, and who have presented over the past 33 years – a living history of our department! I will end by focusing on faculty who have won awards over the last year.

T2) How does interoceptive exposure work in the treatment of anxiety sensitivity?

Sherry H. Stewart, Brigitte Sabourin, Margo C. Watt,³ Jaye Wald,⁴ & Maureen Balcom

³St. Francis Xavier University, Department of Psychology; ⁴University of British Columbia, Department of Psychiatry

Anxiety sensitivity (AS) is implicated in the development/maintenance of anxiety disorders. A brief cognitive behavioural (CB) intervention is effective in decreasing fear of anxiety in high AS women (Watt et al., 2006). This process-based study explored the specific role of the interoceptive exposure component (i.e., running) in explaining intervention efficacy. Female undergraduates, selected as either high or low in AS, participated in three 1-hour CB group sessions that included the running component on the third day. Participants also completed ten 10-minute running sessions between the third day of therapy and the 10-week follow-up. Running was effective in increasing heart rate. Compared to low AS participants, high AS participants displayed greater somatic, affective, and cognitive reactions to running, and showed a greater decrease in affective and cognitive reactions over sessions. These preliminary results suggest that interoceptive exposure may exert its effects in decreasing AS by reducing catastrophic thinking about arousal-related sensations.

Note: This talk will be jointly presented by Sherry Stewart and Brigitte Sabourin.

T3) Preserving Our History with Digital Video: The Snapshot Project

Ron Hoffman & Heather Schellinck

Video records documenting the history of the Dalhousie Psychology Department are few in number, and in too many cases rapidly deteriorating. The old analog recordings on magnetic tape will be a difficult challenge to recover. On the other hand, modern digital video recording technology holds promise for increased longevity of archival records, and offers relative ease in recording. In the winter term this year, students in the Psychology/Neuroscience 3010 class began a project to record on digital media a video "snapshot" of the department as it exists today, by interviewing faculty members on camera and then producing a short edited version of each interview. In this presentation, we will describe the project and the training given the students, and show some examples of the resulting material that will go into the departmental archives.

T4) Two mechanisms underlying Inhibition of Return

Ana B. Chica, Tracy L. Taylor, Juan Lupiáñez¹ & Raymond M. Klein

¹ University of Granada, Spain

Inhibition of Return (IOR) refers to slower reaction times to respond to targets presented at previously stimulated, inspected, or attended locations (Klein, 2000). Taylor (1999; Taylor & Klein, 2000) showed that IOR can affect both input and output processing, depending on whether the oculomotor system was in a quiescent or in a prepared state, respectively. The motoric flavour of IOR was revealed by the finding that IOR delayed spatial responses even when there was no possibility of an effect on input processing (because there was no target at the location that was supposedly inhibited; the target was a centrally presented arrow). That the “motoric” flavour of IOR did not affect input processing was implied, but not directly demonstrated by Taylor’s experiment. If the motoric flavour of IOR is truly non-perceptual and non-attentional, then targets presented at supposedly inhibited locations should not show evidence of this inhibition when the responses to these targets are not spatial. We combed the literature for any studies in which IOR was generated by an eye movement and its effects tested with a non-spatial discrimination task. Finding none, we conducted such a test. Strongly confirming the implication of Taylor’s findings, we found that when the eyes move to the peripheral cue and back to the centre before the target appears (to emphasize the motoric component), IOR is generated in a detection task (in which motor preparation is important) but not in a colour discrimination task. The same colour discrimination task showed IOR when eye movements to the cue were prevented (and thus the motoric component of IOR was not activated).

T5) Manipulability Affects the Time to Name Objects when they are in Graspable Orientations

Joshua Salmon & Dr. Patricia McMullen

Previous research has shown that during object naming tasks, participants are slower to name objects they could normally pick up with one hand (manipulable) compared to objects they could not pick up. The current research explores whether this effect of manipulability on object naming is sensitive to the ability to grasp the object as assessed by its orientation. Participants were asked to name both manipulable and nonmanipulable objects shown in three different orientations (rotated 0°, 60°, and 300° clockwise). Preliminary results with twelve right-handers replicate previous research with participants being slower to name manipulable objects and slower to name objects rotated in the picture plane. Further, these results suggest an interaction, with the effect of manipulability bigger at graspable orientations (300° relative to 60° degrees). This interaction appears to be strongest for the first block of trials, and disappears with

practice. These results confirm that effects of manipulability on object naming are related to their ability to be grasped.

T6) Sex Differences in Treatment Outcomes of Patients with Comorbid Depression and Substance Use Disorders

Brigitte Sabourin, Dr. Sherry H. Stewart, Dr. Margo C. Watt

Substance use disorders (SUD) and depression are two prevalent and highly comorbid disorders. It is believed that comorbid depression in women with SUD's tends to be more independent of the SUD and less likely to remit following SUD treatment than comorbid depression in men with SUD's. On the other hand, there remains much uncertainty regarding prognosis of the SUD for individuals with comorbid depression, with studies showing either worse, equal, or better treatment outcomes for comorbid versus non-comorbid individuals. As part of the present study, clients entering treatment at Addiction Prevention and Treatment Services in Capital Health will be assessed for SUD's and depression upon entering treatment and again after three to four months of treatment. Sex differences in rates and in the presentation of comorbid depression will be determined. The study also aims to compare SUD treatment outcomes for women and men with and without comorbid depression, as well as to assess depression outcome after SUD treatment.

T7) Genetics of Early Psychosis and the Influencing Effects of Adolescent Drug Use

Heather Fulton, Sean Barrett and Sherry Stewart

Many argue that schizophrenia develops through a gene by environment interaction. Interestingly, most individuals with schizophrenia abuse drugs making drug use a possible environmental trigger. More specifically, when individuals with an underlying genetic vulnerability use certain drugs at a critical developmental window, an interaction may occur where they develop the disorder. A recent study by Caspi et al. (2005) implicated a particular genetic polymorphism; individuals who had this polymorphism and used cannabis in adolescence were at five-fold increased risk for the development of schizophrenia. However, the investigators did not control for use of several other substances (e.g., tobacco). Thus, the results may not be specific to cannabis. This study will investigate the role of genes in NS Early Psychosis Program patients and matched control participants, and then relate this information to each individual's substance use history to test the gene x environment interaction theory and its specificity to cannabis.

T8) Impact of Methylphenidate on Sleep in Children with ADHD

Penny Corkum

Introduction: Children with ADHD are five times more likely to have sleep problems than typically developing children. Sleep problems can cause serious behavioural, learning/cognitive, emotional and social impairments. Our previous research has suggested that the primary treatment for ADHD (i.e., stimulant medications such as Ritalin) may be one of the underlying causes of sleep disturbances in children with ADHD.

Methods: Participants (n=28) were diagnosed with ADHD based on an assessment conducted in a multisystem, multidisciplinary ADHD Clinic. All children from this clinic who were diagnosed with ADHD and were considered appropriate for a trial of stimulant medication were recruited to participate in the study. Sleep was measured using subjective (questionnaires, sleep diaries) and objective (actigraphs) measures during a one week baseline period and a three week randomized stimulant medication trial (placebo, low dose, moderate dose).

Results: Results of repeated measures ANOVA indicated that sleep was negatively affected by stimulant medication. Based on actigraphy, the average change in sleep duration was a decrease of 63 minutes and sleep onset increased by an average of 40 minutes. However, the number of night time awakenings decreased and overall sleep efficiency was not affected. Similar results were found based on the sleep diary data, but not the questionnaire data.

Conclusions: Children with ADHD had typical sleep before starting stimulant medication, however, during the acute medication trial there was a statistically and clinically significant change in their sleep. There was some evidence of increased consolidation of sleep, perhaps a compensatory mechanism for the sleep disruption caused by stimulant medication. Further prospective research examining both the short and long term implications of stimulant medication on sleep in children with ADHD is needed.

T9) Self-reported sleep difficulties in a community-based sample of children and adolescents with recurrent pain

Nicolle Vincent, Christine Chambers, Penny Corkum, Ben Rusak

Objectives: In clinical samples of children and adolescents who experience recurrent pain, there is evidence of more disturbed sleep compared to those without chronic pain complaints. This study examines self-reported sleep difficulties in a community-based sample with recurrent pain in comparison to a pain-free control group. Possible mediating variables are also explored. **Methods:** Participants were ninety children and adolescents, 8-15 years of age, who experienced recurrent pain or were healthy and pain free. Participants completed self-report measures of their sleep, pain, functional disability, depression, coping style, and family functioning. **Results:** Participants with

recurrent pain reported significantly more sleep difficulties than did those in the pain-free control group. There was a significant positive association between increased reports of sleep difficulties and higher scores for pain (frequency, duration, intensity). Further analyses will be conducted to identify possible mediating variables.

Conclusions: These results suggest that sleep disturbance is associated with pain in a community-based sample of children and adolescents with recurrent pain.

T10) Effects of Continuous Positive Airway Pressure (CPAP) Treatment of Obstructive Sleep Apnea on Daily Functions, Mood and Quality of

Lau E.Y.Y. , Eskes G.A., Morrison D.L., Rajda, M., Spurr K.F.

Introduction: Obstructive sleep apnea (OSA) is characterized by disrupted breathing during sleep, leading to hypoxemia and fragmented sleep at night, daytime sleepiness, and changes in cognition and mood. Here, we report preliminary analyses of an ongoing study of the effects of CPAP treatment on sleepiness, sleep quality, daily functions, mood, and quality of life (QoL). Methods: Thirty CPAP-treated patients with moderate to severe OSA were studied with an overnight polysomnographic sleep study, measures of sleepiness and sleep quality, mood, functional outcomes, cognition, and QoL. Pre-treatment respiratory disturbance index (RDI), mean oxygen saturation (SpO₂), sleep quality, and sleepiness, and retrospective ratings on functional outcomes and QoL were also available. Results: CPAP improved breathing disruptions, sleep quality, daytime sleepiness, functional outcomes, and QoL. Post-treatment mood, functional outcomes, and quality of life are predicted by sleepiness and sleep quality. Conclusion: CPAP is an effective treatment for OSA in ameliorating breathing disruption and oxygen desaturation during sleep. However, there are residual difficulties in daytime outcomes associated with sleepiness and sleep quality.

T11) Sleep is the Best Remedy for Sleep Loss, But How are our Hormones Affecting our Ability to Recover Sleep? —A Case for Females

Samüel Deurveilher¹, Michael Wilkinson^{2,3}, Benjamin Rusak^{4,5,6} & Kazue Semba^{1,4} *Departments of ¹Anatomy & Neurobiology, ²Physiology & Biophysics, ³Obstetrics & Gynaecology, ⁴Psychology, ⁵Psychiatry, ⁶Pharmacology; Dalhousie University*

We investigated the effects of ovariectomy with or without estrogen and/or progesterone replacement on spontaneous sleep and recovery sleep after 6 h of sleep deprivation in female rats. Male rats were also included for comparison. Females were ovariectomized and implanted subcutaneously with capsules containing oil, estrogen, progesterone or both hormones. Males received oil-filled capsules. All were implanted with EEG and EMG electrodes. During the baseline recording, ovariectomized rats without hormone replacement tended to spend more time in both non-rapid eye movement (NREM) and REM sleep than those with estrogen and/or progesterone replacement and males. Following 6-hour sleep deprivation, ovariectomized rats without hormone replacement were much slower in recovering REM sleep (but not NREM sleep) than the other groups, whereas ovariectomized rats with the highest dose of estrogen showed the lowest NREM EEG delta power (a measure of sleep intensity). These results indicate

that ovarian hormones can significantly affect REM sleep and NREM delta power recovery in female rats.

T12) The Role of Caffeine Exposure and Dependence in a Conditioned Flavor Avoidance Paradigm

Sarah E. Dreumont-Boudreau & Vincent M. LoLordo

Two experiments assessed the value of a flavor that signaled 24 hrs of caffeine exposure (Signal CS) in male Sprague-Dawley rats. In Experiment 1, naïve rats were given a Kool-Aid flavor followed by 24 hr access to caffeine. Several days later, they were given a different flavor followed by water (Neutral CS). When given a choice between the two flavors, rats avoided the Signal CS and consumed more of the Neutral CS. Avoidance of the Signal CS was maintained even after the rats were made dependent on caffeine. In Experiment 2, Group 1 was maintained on 21 days chronic caffeine, while Group 2 was maintained on water. Both groups were then given a Signal CS followed by 24 hrs of caffeine, and several days later, a Neutral CS followed by water. Group 2 avoided the Signal CS, and consumed more of the Neutral CS. However, Group 1 consumed equivalent amounts of both the Signal and Neutral CSs.

T13) Caffeine dependence in rats: Effects of dose and exposure on conditioned flavour avoidance.

Rachel Dingle & Vincent M. LoLordo

In order to examine the parameters of oral caffeine dependence, groups of rats were chronically exposed to caffeine-tainted water (1.0 g/L) for 5, 10, 15 and 20 days. Upon removal of the caffeine, rats were exposed to one of two novel Kool-Aid flavours (CS Withdrawal). Twelve days later, rats were exposed to the second Kool-Aid flavour (CS Neutral). A 24-hour 2-bottle test showed that rats that had been exposed to caffeine for 20 days, but not 5, 10 or 15 days, strongly preferred the CS Neutral. In a second experiment, groups of rats were given caffeine-tainted water for 21 days at one of four concentrations (1.0 g/L, 0.5 g/L, 0.25 g/L, and 0.125 g/L), after which withdrawal and neutral CSs were established. Only rats that drank the 1.0 g/L caffeine solution strongly preferred the CS Neutral. This suggests that long exposure to a strong caffeine solution is required in order to induce dependence in rats such that a CS associated with the withdrawal of caffeine becomes avoided.

T14) The Effect of Consumption of Kola Nut (*Cola Nitida*) Diet on Locomotor Behaviour and Anxiety in Mice

Sunday A. Bisong, Gideon Oyibo, Eme E. Osim

Department of Physiology, College of Medical Sciences, University of Calabar

Kola nuts are eaten in Nigeria (mostly in the Northern part) during ceremonies and as stimulants. The long term effect on behaviour is yet known. Following chronic consumption (4 weeks feeding) of kola nut (*cola nitida*) diet (50%w/w), patterns of locomotor behaviour and anxiety were studied in 16 male Swiss white mice (18-32g body weight), using the open field, elevated plus maze and the light/dark transition box. Mice in the control group (n = 8) were fed normal rodent chow while those in the kola group (n = 8) were fed kola diet (50% kola, 50% rodent chow) for 4 weeks. All animals were allowed free access to clean drinking water. The frequency of line crossing in the open did not differ between the kola and control groups. The frequency of rearing against the wall did not differ, but the frequency of rearing in the open field was lower in the kola group ($p < 0.05$). Grooming and genital licking were also lower in the kola group compared to control ($p < 0.001$, 0.05 respectively). In the elevated plus maze the duration in the open arm in the kola group was higher than the control group ($p < 0.01$). The frequency and duration of head dips did not differ. The frequency and duration in the closed arm were lower in the kola group compared to control ($p < 0.05$). The duration of grooming, genital licking in the kola group was higher in the closed arm compared with control ($p < 0.01$). The kola fed animals spent more time in the light region of the light/dark transition box ($p < 0.01$), showed more rearing ($p < 0.05$), and spent less time in the dark region when compared with controls. These results indicate that consumption of kola nut diet decreases anxiety-related behaviour in mice in the elevated plus maze and the light/dark transition box but does not affect locomotor activity in the open field.

T15) Anxiety and Risk-Taking in a Mouse Model of Sensation Seeking

Jennifer Stamp, Danny Mendelsohn, Michael Wong, David Cyr, & Heather Schellinck

Sensation-seeking is a personality trait characterized by the desire to experience novelty. Sensation seeking is typically associated with low anxiety, high risk-taking and vulnerability to drug abuse. One potential animal model of human sensation-seeking is high response to novelty in rats. High responders (HRs) choose novel environment over a familiar one and show an increase in locomotor activity in the new environment. These rats also display lower levels of anxiety and an enhanced propensity to self-administer drugs of abuse. This project had two main aims: 1) to determine whether the low anxiety in HR rats is also demonstrable in C57 mice and 2) to use two tests of anxiety (open field and elevated plus maze) as well as a test to specifically assess the dimension of risk taking (predator exposure). HR mice were distinguished from their LR counterparts based upon the amount of activity in an open field. In the open field, HRs spent more time in the “risky” center and less time in the “safe” corners compared to LRs. There were no significant differences between HRs and LRs in any of the measures in the elevated-plus maze but HR mice spent significantly more time than LRs in an area of a predator arena which was in proximity to a food-deprived rat. These data suggest that attributes of human sensation-seeking are demonstrable in mice.

T16) What route do visual currents take in generating the fly’s electroretinogram (ERG)?

Steve Shaw

The ERG is an easily recorded, generic visual response, neglected since the 1960s but recently resurrected as an assay tool for investigating the molecular biology of synaptic transmission in *Drosophila*. Using the blowfly *Calliphora*, I’m attempting to determine the real origin of the ERG, mistakenly believed to have been solved long ago.

Transient ERG currents generated in the first synaptic zone add arithmetically to those generated by the retina, but the direction of current flow uncovered here is unconventional. Currents move out from the synapse into the retina, surprisingly so as the retina appears to be a cul-de-sac surrounded by impervious cuticle. That the cornea is actually a good conductor has been confirmed by tracking responses over it with a new fangled surface electrode, after current has been injected alternately on the cornea then just inside it. ERG currents behaved similarly: the cornea has low internal resistance, though a barrier exists at its intersection with head cuticle.

By tracking the angular response of ERG components over $\sim 250^\circ$, ERG currents have been found to enter the cornea only where the light source illuminates active neurons underneath. Some of this current leaves again from ‘inactive’, unilluminated zones, while the rest loops out around the eye, giving rise to the recorded ERG.

T17) Plasticity of microcircuits in the insect central nervous system

Claudia Groh and Ian Meinertzhagen

The goal of our project is to study cellular components of neuronal plasticity and their outcomes in the behavioural plasticity of two model insect species (the honeybee *Apis mellifera* and the fruit fly *Drosophila melanogaster*). In these species, synaptic complexes within the first optic neuropil and within the synaptic input region of the mushroom body (a higher sensory integration centre required for olfactory learning and memory) facilitate the investigation of fundamental principles of neuronal structural plasticity. My previous work revealed specific, environmentally induced and age-dependent structural plasticity within the brain of female honeybees, but there is still little information about the underlying cellular, subcellular and molecular events underlying such changes. Using electron microscopy and immunofluorescence labelling of filamentous (f-) actin, which is enriched in dendrites, my proposed project seeks to narrow the gap between two hypotheses that: a) developmental and adult plasticity of these synaptic circuits in the insect brain may underlie behavioural plasticity; and b) the source of dendritic structural plasticity is based on the dynamics of motile f-actin.

T18) Synaptic circuits formed by input neurons to the fly's optic medulla

Shini-ya Takemura, Zhiyuan Lu and Ian Meinertzhagen

The visual system of a fly is intricate, precise, and supports rapid vision. Photoreceptor inputs are received at the two outermost neuropiles of the optic lobe. Many classes of optic lobe neuron, each with a unique morphological signature, are reported from Golgi impregnation. Their synaptic circuits are known completely in the first neuropile, or lamina, but remain a formidable challenge in the second, the medulla. The shape of a neuron is only a guide to how it may connect with other neurons, which must ultimately be investigated by electron microscopy (EM). Most synaptic sites are divergent dyads, and much of the neuropile comprises interneurons and their local microcircuits. Using: a) unlabelled EM series of the distal medulla; and b) the Gal4/UAS system to label specific neurons at EM level, we are analysing synaptic connections of lamina input terminals. Spectral sensitivity is encoded by photoreceptors R7 and R8, and motion sensitivity by lamina cells L1 and L2. Remarkably, R8 provides input to R7 and also to L1, structural evidence for a pathway between colour and motion-sensitive channels.

T19) Immunocytochemical localization of synaptic and other proteins in the photoreceptor terminals of the fruit fly, *Drosophila melanogaster*.

Yoshitaka Hamanaka and Ian Meinertzhagen

The fruit fly, *Drosophila melanogaster*, is a useful model organism to study the localization of the proteins that contribute to synaptic function because it is genetically manipulable and because its neurons have a distinctive synaptic ultrastructure that includes a specialized pre-synaptic organelle at the release site, T-bar ribbon. The latter comprises a platform surmounting a pedestal. I examined the protein composition of the presynaptic T-bar ribbon and its associated synaptic organelles, at photoreceptor terminals of *D. melanogaster*. After validating candidate antibodies by confocal microscopy, pre-embedding immuno-EM methods showed that four monoclonal antibodies all clearly label distinct synaptic structures: nc82 (Bruchpilot) and SUK4 (which labels a kinesin epitope) label the T-bar ribbon, nc82 at the edge of the platform, and SUK4 close to its pedestal; anti-discs large recognizes the photoreceptor head of capitate projections that invaginate from surrounding glia; while aa2/2 (raised against a homogenate of *D. melanogaster* brains) localizes to the synaptic vesicle pool.

T20) Ectopic synapses in the visual system of the fruit fly *Drosophila melanogaster*

Tara Edwards and Ian Meinertzhagen

The visual system of the fruit fly is remarkable for its numerical and spatial determinacy, especially at its identified photoreceptor synapses. In *Drosophila* we can examine ectopic synaptogenesis by genetic means, by targeting photoreceptor axons to an incorrect neuropile in the brain. Photoreceptors are of two types, R1-R6 that innervate the first optic neuropile, the lamina, and R7-R8 that innervate the second, the medulla. Target cells within the lamina persist and maintain expression of cell-specific markers, even when their normal R1-R6 photoreceptor inputs terminate in the medulla. Analysis of lamina ultrastructure reveals that the axons of these mis-targeted photoreceptors, even though they penetrate the lamina, nevertheless form reciprocal synaptic inputs with their normal lamina targets. In addition, these mis-targeted R1-R6 photoreceptors also form ectopic synapses within the medulla, thus providing input to two neuropiles. A subset of these ectopic synapses are characteristic of photoreceptor synapses in the lamina, tetrads with four postsynaptic partners, which are otherwise not found in the R7-R8 photoreceptors that normally terminate within the medulla.

T21) The influence of heavy metals on the morphology and plasticity of neurons in the fly's visual system

Joanna Borowska and Ian Meinertzhagen

The insect's brain provides an excellent, expendable model to study basic processes occurring in neurons. Using the housefly's brain, I have previously examined the toxic effects of heavy metals (lead, cadmium, zinc and copper) on neuronal morphology and plasticity. My analysis of neurotoxicity concentrated on the influence of these metals on the plasticity of two classes of monopolar cell, L1 and L2, in the lamina of the optic lobe. Changes in the volume of these interneurons depended on the type of metal and its concentration, and revealed that heavy metals penetrate the haemolymph-brain barrier. Rare genetic errors in metal metabolism, or accumulation of metals through environmental or occupational over-exposure can also elevate brain metals to cause neurodegeneration. When they degenerate, neurons take up heavy metals, probably because fail to pump them. With workers at the Canadian Light Source, we will use X-Ray absorption spectroscopy to image the metal profile of fly brains undergoing neural degeneration. For this, we are examining a fly model of Alzheimer's Disease, mutant for the human gene for human A β 42 amyloid peptides, a major component of senile plaques, and are developing methods to image the brains of these flies non-invasively, as a model for the progression of neurodegenerative disease in humans.

T22) *Drosophila* mutants *white*, *scarlet* and *brown* differ not only in their eye colour but also in the distribution and head content of biogenic amines

Janusz Borycz, Jola Borycz, Rita Kostyleva and Ian Meinertzhagen

The *white*, *scarlet* and *brown* genes in *Drosophila melanogaster* are named because their mutants have these abnormal eye colours. *white* mutant flies also have a behavioural phenotype that indicates a neural action. The genes encode ABC transporters involved with the uptake and storage of precursors of pigment granules. We find additional effects on head biogenic amines: all three have about half the normal head histamine (HA), a neurotransmitter at photoreceptor terminals, as well as reduced 5-HT; and *white* also has reduced dopamine (DA).

The mutants also have abnormal cellular distributions of HA, 5-HT and DA. In centrifugates of wild-type brain homogenates, a vesicle-enriched pellet fraction contains ~75% of head HA with 25% in the supernatant. In *white*, *brown* and *scarlet* these proportions are reversed. Similar determinations reveal a reversed ratio between supernatant and pellet fractions for 5-HT and DA as well, and mutant photoreceptor terminals have reduced numbers of synaptic vesicles. We suggest that these three genes may influence the intravesicular contents of the neurotransmitters histamine, serotonin and dopamine in neurons, so as to affect transmission by these neurons.

T23) *Drosophila* synapses: protein complexes viewed using EM tomography

Birgit Greiner, Nicole Delaney, Nancy Butcher, Rorbert M. Marshall and Ian A. Meinertzhagen

Genes programme the physiology of synapses by coding for the synaptic proteins that regulate synaptic function. However, conventional electron microscopy of thin sections is unable to visualise the organisation of such protein complexes because these overlap each other within the depth (~50nm) of an ultrathin section. Recent developments in EM tomography solve the problem of overlap, deriving depth information from wide-angle image tilt series, and revealing structure at the sub-nanometre resolution inherent in an electron beam. Using EM tomography methods we have reconstructed the T-bar ribbons and other synaptic organelles at the release sites of two types of *Drosophila* wild-type synapse, the neuromuscular junction and photoreceptor tetrad synapses of the fly's lamina. These reconstructions provide a basis against which we will compare structural perturbations in synapses of flies mutant for the genes, such as *endophilin* or *bruchpilot*, coding for synaptic proteins.

T24) That's what Friends are For: A Comparison of the Qualities of Single- and Dual-Perpetrator Murders

Stephen Porter, Michael T. Woodworth and Marcus Juodis

To date, there has been no investigation of possible qualitative differences between homicides committed by perpetrators acting alone versus those who acted with an accomplice. In this presentation, the findings from the first empirical study of single-versus dual-perpetrator homicides will be described. This study involves a thorough examination of the official crime descriptions of 124 Canadian homicides, 84 involving a single perpetrator and 40 involving two perpetrators. The crimes are being compared in terms of perpetrator motive, instrumental/ reactive violence, sadistic violence, gratuitous violence, sexual elements, and victim characteristics. Preliminary findings suggest that dual-perpetrator homicides are associated with a higher level of instrumental aggression (i.e., premeditation, external gain) than single-perpetrator homicides. Further, murders by single perpetrators were significantly more likely to display sadistic and sexual violence than murders by dual-perpetrators. Single perpetrators were more likely to target female victims, whereas dual perpetrators targeted both males and females.

T25) Conflict in Dating Relationships

Kate Thompson

The present study assessed the relationship between dating abuse and irrational thinking and the hostile attributional bias. This was done using the Conflict in Adolescent Dating Relationships Inventory (CADRI) as the measure of past abusive behavior, the Survey of Personal Beliefs (SPB) as the measure of irrationality, and the rating of a video depicting an ambiguous interaction between two triangles as a measure of the hostile attributional bias. It was hypothesized that greater levels of abuse would be significantly related to higher levels of irrational thinking, and greater incidence of the hostile attributional bias. Hierarchical regression analysis indicated a small, but significant, relationship between abuse and irrational thinking (particularly, the inability to tolerate frustration). Limitations and future directions are discussed.

T26) A collaborative, community based exploration of how post-traumatic chronic arousal acts as a mediator between exposure to violence and subsequent alcohol use in a sample of school attending Mi'kmaq adolescents

Marc Zahradnik, Sherry H. Stewart, Doreen Stevens, Christine Wekerle, & Christopher Musquash,

This study is part of a larger ongoing collaborative mental health initiative between our lab and a Nova Scotian Mi'kmaq community. Given the strong relationship that post-traumatic stress (PTS) symptoms have with both childhood exposure to violence (EV) and substance misuse, particularly alcohol, we hypothesized that—based on the “self-medication” hypothesis—in a sample of school attending Mi'kmaq adolescents, PTS symptoms will mediate the relationship between EV and subsequent alcohol use (AU). From a total sample of 166 (52% female, 42% male, & 6% unknown) adolescents (minimum age 14), individual regression analyses (Baron & Kenny, 1986) yielded empirical support for chronic arousal PTS symptoms, total PTS symptoms, and depressive symptoms as “partial mediating” variables that help explain the relationship of EV with later AU. However, a regression analysis that included all relevant predictors—EV, each of the 3 PTS symptoms, total PTS symptoms, and depressive symptoms—resulted in a final model that accounts for 28% of the variance in predicting AU, and in which only the chronic arousal symptom cluster of PTS and EV remained significant.

T27) Incidental forgetting as a by-product of intentional forgetting

Jonathan Fawcett and Tracy L. Taylor

The present study explored the consequences of item-method directed forgetting on incidental learning of probe words. During the study phase, study words were presented individually followed by an instruction to remember (R) or forget (F). Pink and blue probe words were presented at stimulus onset asynchronies (SOAs) of 1400, 1800 or 2600 ms in relation to each memory instruction; participants were required to make a speeded colour discrimination of each probe word. After the study trials, a yes-no recognition task measured retention of R and F study words and incidental learning of probe words. Recognition data indicated a significant directed forgetting effect, with greater recognition of R than F study words. Importantly, F instructions increased discrimination RTs and impeded later recognition of probe words presented at the earliest SOA. These findings demonstrate that intentional forgetting interferes with the formation of incidental memories. These results are inconsistent with the dominant hypothesis in the literature, which suggests that forgetting is accomplished by allowing F items to passively decay from working memory.

T28) Temporal Order Judgement in Regards to First Base Plays in Baseball

Kristin G. Smith

Sports fans, players and coaches can get very heated about important games, often disagreeing with the umpire's calls. Basic research on the effect of attention on temporal order judgments suggest that these disagreements may occur because the spectators and umpires are attending different regions of space. We sought to determine if attending to different regions in space would cause a difference in the decision of an individual to call a play "safe" or "out". In E1 participants were instructed to attend to the ball or the foot. The results showed that attention mattered: when attending the foot there were more "safe" calls than when attending the ball. In E2, when participants were instructed to attend to the glove or the base, no significant effect of attention was found. It was hypothesized that the participants may have been able to attend both the base and the glove due to their proximity. In E3 participants were instructed to attend both stationary objects, or both moving objects. It was found that the data was similar to that of E2 and there was no significant difference between the attend stationary and the attend moving conditions. Across all trials there was a bias to call "safe" more often than "out".

T29) Psychophysical Quantification of Configural Face Processing Using Just Noticeable Differences

Heath Matheson and Paddy McMullen

To quantify the degree to which upright faces are processed with a 'configural processor' relative to inverted faces and objects, psychophysical, just-noticeable differences (jnds) were determined for displacement of critical features (e.g. eyes, windows) within upright and inverted faces and houses and other objects. Stimuli were closely matched on low-level visual features of size, luminance, and shape. Observers were most sensitive to displacement of critical features in upright faces. Sensitivity was equivalent for inverted faces, upright houses and inverted houses. This technique provides quantified differences in the degree to which configural processing is used.

T30) Low-Level Processes Contribute to the Production of the Inhibition of Return Effect

Kristie Dukewich (Dalhousie) and Susan Boehnke (Queen's)

The ability to detect and orient to unexpected novel changes in our environment is one of the primary functions of attention. Inhibition of return (IOR) is thought to be a functional after-effect of visuo-spatial orienting, defined as an increase in response times to a target when it is presented at a previously stimulated location compared to when it appears at a new location. Recently it was shown that IOR is correlated with the decrement in the visual response of superior colliculus neurons to a target after responding to a cue at the same location (Dorris et al, 2002). Furthermore, for a sequence of visual stimuli, a smaller visual response occurs with each subsequent stimulus presentation, likely through a process of adaptation and/or habituation (Boehnke et al, 2006). This suggests that the IOR effect should grow as a function of the number of times a cue is presented prior to the target. To investigate this, we used multiple cue presentations (blocked by number of cues) and multiple cue identities in a Posner cueing task requiring a manual response to a target. The results of the experiments demonstrate that habituation and/or adaptation can occur based on a stimulus' location and affects the timing of subsequent behavior, providing further evidence that low-level sensory processes play a role in producing the IOR effect.

T31) Black and White Like Me

John Barresi

John Griffin's classic on racism, *Black Like Me*, provides an interesting text in which to investigate the development of a dialogical self. Griffin becomes a black man for only a short period of time, but during that time he develops a black social identity that contrasts radically with his former white identity. He develops a schizoid sense of self. He is white and he is black. When he looks into a mirror on several occasions he engages in a dialogue with himself, as both a black and a white person. At first these two identities are so different that there is no "sympathy" between them. But through his experience, he eventually overcomes the dichotomy of two opposing selves, and acquires a personal identity, neither white nor black, but just human. In this talk, I will trace the development of these dialogical selves and the emergence of this new human identity.

T32) The development of 'third-party' gaze following

Chris Moore & Cherie Collicott

By 9 months of age, infants will typically follow the gaze of an interactive partner to a target outside of the immediate visual field. We know from recent research (e.g. Floor & Akhtar, 2006) that infants are only able to learn words and actions in purely observational (or 'third party') contexts from about 18 months. In this study we examined when infants would follow the gaze of someone with whom they are not interacting. Infants of 12 and 18 months observed an interaction between their mothers and an experimenter during which the two adults periodically turned to look at an interesting sight. Coding of infants' tendency to follow the adults' gaze showed that whereas 12-month-olds did not follow gaze, 18-month-olds did. This difference was not due to age-related differences in the tendency to look at the adults. These results are consistent with the idea that young infants learn best in interactive joint attentional settings but by the end of infancy can start to learn purely through observation.

T33) THE RUBBER HAND ILLUSION: Intermodal Perception and Self-Ownership in Adults

Timothy Dummer

Mental representation of the body arises from the intricate relationship between intermodal sensory input and a continually changing conceptual diagram. Any interference to this process, such as when a person views a rubber hand being stimulated in synchrony with the occluded real hand, will lead to a misperception as tactile sensations are falsely referred to the non-body part. Our research is focused on further analyzing the nature of the rubber hand illusion using conditions of intentional (goal directed) and random movement. We predict that intentional movement will generate a stronger rubber hand illusion based on previous research that links volition with body perception. Results failed to produce a robust rubber hand illusion but

revealed a possible attentional component to the process of body identification. In the present study we plan to replicate the previous experiment with the inclusion of a goal-directed movement condition and measure of attention.

T34) I feel the same way: Recognizing facial emotion as appropriate to context.

Cindy Hamon-Hill, John Barresi

Almost all research in emotion recognition to date focuses on categorizing exaggerated facial displays of primary emotions. Little research has focused on how we recognize more complex emotional expressions in their natural contexts - in particular in contexts in which we share emotions with others but do not explicitly label them. The main purpose of our current project is to develop a paradigm in which natural emotional expression to given contexts are used in research on emotion recognition. We focus on the ability to recognize when someone is sharing an emotional reaction similar to self with respect to a stimulus context, rather than on the more cognitive/linguistic task of labelling that emotion. We expect that the development of such a paradigm will open up new ground for investigating the basic processes involved in understanding shared emotional states, where articulation through use of emotion category labels is not involved.

T35) Reading between the Lies: An examination of cues to deception in emotional facial expressions

Leanne ten Brinke & Dr. Steve Porter

Since the 1960s, Ekman and colleagues have argued that certain aspects of facial communication are not controllable and that inconsistent emotional expressions, including “micro-expressions”, can betray a deceiver’s true emotion. Despite the widespread popularity of this idea, it has received little empirical attention. This study compared genuine and deceptive (simulated, neutralized, masked) facial expressions. Each 1/30th second frame of the videotaped clips then was coded for the presence and presentation time of the universal emotional expressions in the upper and lower facial regions. Results indicated that inconsistent expressions occurred significantly more frequently in masked than in genuine expressions. Additionally, inconsistent expressions were more frequent and longer in falsified negative (fear, disgust, and sadness) than happiness expressions. Micro-expressions occurred in 21.95% of participants in 2% of all expressions (14 micro-expressions in total), and all involved the upper or lower face only (unlike the traditional “full face” conceptualization). Implications for identifying concealed emotions in forensic and security settings will be discussed.

T36) Parenting Stress and Maternal Depression in Rural Mothers

Deborah Johnson-Emberly

Over the past 10-15 years, a consensus has emerged indicating that the causes of parental stress, dysfunctional parenting and child outcomes are best understood using dynamic multivariate models that acknowledge the interaction between variables and the changes in behaviours and cognitions in the parent and child as a result of recurring interactions (Abidin, 1990). Accepted models of parenting stress encompass child characteristics, parent characteristics, environmental characteristics and parent child interactive stress. Previous research has identified strong links between maternal psychological distress/ functioning and parenting stress, dysfunctional parental behaviours, and the social and emotional well being of the child. The stress of parenting is great for all parents. Unique aspects of rural life can compound the factors of parental stress. These include social isolation, employment issues and a general lack of available resources, such as access to health care and childcare. The connections between these sources of rural stress and parenting stress have not been studied. In addition, the mediating role of maternal depression on the direct pathway from general stress to parenting stress has not been investigated. A conceptual model for understanding these connections between rural experiences of stress, maternal depression, and parenting stress will be discussed in this presentation.

T37) Memory complaints after temporal lobe resection are the exception, not the rule.

Mora Martin¹, Jaime Williams^{2,3}, Shirley Munk² and Jeannette McGlone^{1,2}

Departments of Psychology, Dalhousie University¹, Capital District Health Authority²,
University of Regina³

Temporal lobe resection (TLR) is an efficacious and cost effective treatment for medically intractable complex partial seizures. A risk of mesial TLR is memory loss. Three converging methods asked, "Are patients aware of memory loss in their daily lives one year after surgery?" Method I examined whether 64 TLR patients from Nova Scotia reported group changes in self-

reported and verbal memory ability pre- to post-operatively. Mean verbal memory ability declined significantly after left TLR; however, patients and their relatives rated the patients' memory to be significantly *better* after than before surgery. Method II examined the magnitude of change in subjective ratings for each individual (i.e., Reliable Change Index), based on control data obtained from unoperated epilepsy patients who were medical managed over the year. The majority of TLR patients were classified as unchanged or improved in memory self-ratings whereas fewer than 10% fell into the significant deterioration classification. Method 3 was a meta-analysis that synthesized outcomes of six publications on self-reported memory change pre- to post-op in persons with TLR. Twenty-two effect sizes were calculated based on a total sample of 450 patients and 171 relatives. The weighted mean effect size was +0.25. Hence the consensus is that memory is judged to be significantly *better* following TLR than before. We conclude that objective and subjective estimates of memory are dissociable. Both risks need to be discussed prior to elective surgery.

T38) Does the risk of memory loss influence decision making by physicians and patients who contemplate temporal lobe resection for control of seizures?

Jaime Williams^{1,2} Jordan Green³, and Jeannette McGlone^{1,3}. Capitol District Health Authority, Halifax, NS¹; Department of Psychology, University of Regina², Department of Psychology, Dalhousie University³

Temporal lobe resection (TLR) is a successful, although underutilized treatment for complex partial seizures. One reason for this underutilized may be the perceived risk of severe and permanent memory loss. However, patients' and physicians' perceptions of the risk of memory loss following TLR have not been empirically investigated. In this presentation, we propose a series of studies using a triangulation of methodologies examining whether fear of memory loss contributes to decision-making about elective brain surgery. First, we propose conducting focus groups with non-surgical, pre and post TLR patients to uncover their thoughts and attitudes towards TLR surgery and the risk of memory loss. Second, we will survey epilepsy specialists querying their thoughts regarding memory loss including how they communicate the risk to patients and what role the risk has in recommending and proceeding with TLR. Finally, we will explore information available on the Internet regarding TLR and the risk of memory loss, including the accuracy and consistency of the information. Implications of the research will be discussed including the importance of effective epilepsy-related knowledge transfer and critical comments will be invited.

T39) A multimodal imaging approach for surgical planning in temporal lobe epilepsy

Jodie Gawryluk, Ryan D'Arcy and David Clarke

Functional brain mapping has tremendous potential to guide neurosurgical planning. Preliminary studies have used functional magnetic resonance imaging (fMRI) and event related brain potentials (ERPs) to map the temporal lobe in a healthy control sample and have highlighted the importance of a multimodal imaging approach. The proposed program of research will combine (fMRI) and (ERPs) to focus on resective surgery for temporal lobe epilepsy (TLE). TLE is the most common type of seizure disorder and surgical intervention is often required for medically intractable cases. Functional MRI will allow for a spatial analysis of task related activation (targeting the temporal lobe), whereas ERPs will allow for analysis of specific processes related to that activation. The current program of research consists of two components: 1) fMRI and ERPs will be used to characterize medial and lateral temporal lobe function in TLE patients and healthy controls; 2) pre- and post-surgical comparisons will determine whether multimodal mapping can improve functional outcome.

T40) Interhemispheric transfer of functionally lateralized visual stimuli: An fMRI study at 4T

Erin L. Mazerolle & Ryan C.N. D'Arcy

Split-brain patients are generally unimpaired in daily life, but demonstrate deficits on tasks that require interhemispheric transfer. We studied interhemispheric transfer in healthy individuals using 4-Tesla functional magnetic resonance imaging (fMRI). In addition to visual interhemispheric transfer (examined previously), the experiment was designed to explore transfer of motor information. We hypothesized activation differences as a function of the type of cross (visual or motor). Word and face stimuli were presented to elicit lateralized brain processing in the left and right hemispheres, respectively. Visual transfer was manipulated via hemifield of presentation, and motor transfer was manipulated via response hand. Decreased behavioural performance was observed for crossed conditions. Both visual and motor transfers were associated with activation near and in the corpus callosum, but visual cross activation was more posterior (around the splenium), whereas motor cross activation was more anterior (around the body). This study highlights the benefits of high field fMRI for studying interhemispheric transfer across the corpus callosum.

T41) A framework for novelty detection, pattern completion and memory consolidation in the hippocampus and cortex

Dominic Standage and Thomas Trappenberg

A large volume of data support the hypothesis that the hippocampus rapidly encodes representations that bind distributed cortical memories until cortex can reactivate its own memories. We provide a mechanistic framework for this hypothesis, accounting for novelty detection, pattern completion and memory consolidation in the basal ganglia, hippocampus and cortex. Under this framework, everything subject to attention is encoded by early LTP in the hippocampus, but only novel, goal-directed events are subject to late LTP (I-LTP) in subfield CA1, the site of pattern completion under the framework. Attention and novelty detection emerge from hierarchical, bidirectional cortical processing, corresponding to coherent and incoherent activity respectively. Incoherent activity creates unique hippocampal representations, learned by synapses mediating coherent activity and generating a strong response in CA1. This response converges with goal-directed prefrontal activity in the basal ganglia, leading to dopamine release and I-LTP in CA1. Subsequently, coherent activity regenerates the CA1 representation, activating associated cortical areas. The rate of cortical consolidation depends on interference with existing representations.

T42) Sex, Drugs, and Abrupt Onset Distractors: Made you Look!

Nathan Saruk, Gail Eskes, John Christie, Hilary Duncan, and Ray Klein

Visuo-spatial neglect is a failure to attend, detect or respond to stimuli that appear in contralesional space after a stroke. Neglect is considered to be a disorder of attention resulting from impairment in exogenous control of orienting while endogenous orienting is believed to remain relatively intact. We examined the interaction of exogenous and endogenous control of orienting in an oculomotor capture paradigm in 3 groups: young and older healthy controls, and stroke patients with spatial neglect. Dependent variables included saccadic latency from target onset to saccade initiation and proportion of trials in which the eyes initially moved to an abrupt onset distractor (attention capture) before proceeding to the target. Consistent with previous findings, saccadic latencies were longer in the older group compared to the younger group, but both groups showed equivalent rates of overt attentional capture. In contrast, patients with neglect showed increased capture to right-sided distractors and decreased capture to left-sided distractors.

T43) Of guns and geese: A general review of the 'weapon focus' literature

Jonathan Fawcett

A frequently cited factor in the accuracy of eyewitness testimony is weapon focus, defined as a deficit in later memory for an event or visual scene containing a weapon. Weapon focus has

been attributed to interference from the arousal/threat created by the weapon (arousal/threat hypothesis) or by attentional focus on the weapon (unusual item hypothesis). Laboratory, simulation and field studies are separately reviewed, with emphasis on their implications for both of these hypotheses. It is concluded that weapon focus is a small effect observed primarily in laboratory studies, and not easily generalized to real-world scenarios. This talk is a prototype for a symposium presentation submitted to BBCS and a review article in preparation; therefore, questions or comments are encouraged.

T44) The Role of Self-regulation in Pro-environmental Behaviour

Meredith Flannery

A major challenge in the encouragement of pro-environmental behaviours is demonstrating long-term maintenance of desired actions. Recent research suggests long-term health-related behaviour change may be linked to self-regulation skills (Hall et al., 2006). Like health behaviour, establishing new pro-environmental behaviours often involves problem solving and long-term planning that require self-regulation skills. The proposed study will examine the relation between self-regulation skills and pro-environmental behaviour in adults. Using Ajzen's Theory of Planned Behaviour, self-regulation skills will be assessed to help examine the degree to which "actual behavioural control" explains differences in pro-environmental behaviours at the individual level. Three dimensions of self-regulation will be examined according to a problem-solving framework in the literature on early child development of executive function: problem representation, planning, and execution. It is expected that individuals with better self-regulation skills will show more pro-environmental behaviours. Additionally, actual behavioural control should help to explain significant variance in pro-environmental behaviour.

T45) Is a hands-free phone safer than a handheld phone?

Yoko Ishigami

The nature of the differences in driving-related performance between hands-free (HF) and handheld (HH) phone drivers is not well understood. One purpose of this review is to evaluate the effects of the phone type on driving (-related) performance across different levels of driving fidelity in the published literature. Talking on a phone, regardless of the phone type, has negative impacts on driving (-related) performance especially in detecting and identifying events. Moreover, the HF phones were rarely found to be necessarily safer than the HH phones. These patterns are similar across different levels of fidelity in studies.

T46) Stress Responding and Adolescent Development in a Rat Model System: Alley cats and hood rats

Lisa Wright, Kimberly Hébert, Katherine Muir, and Tara Perrot-Sinal

Recent studies demonstrate that the hypothalamic-pituitary-adrenal axis can be 'programmed' by experiential factors in early life, tying mature stress response output to past environmental conditions. Mesocorticolimbic dopamine function also contributes significantly to adult stress response regulation, and this system undergoes major developmental modifications during the adolescent period. In two experiments, we manipulated levels of adolescent stress using repeated exposure of rats to ethologically relevant cat odour stimuli (periadolescent predator odour; PPO) or a control condition. In Experiment 1, adolescent rats showed a clear avoidance response upon initial exposure to the cat odour. By the final PPO exposure, however, they showed several measures of hyperactivity and spent significantly more time near the odour relative to control rats. Furthermore, long-term effects on defensive behaviours persisted into adulthood, and adult expression levels of the D2 dopamine receptor were lower in prefrontal regions. In Experiment 2 specific sensitivity of the adolescent period was examined by comparing outcomes of PPO with identical manipulations administered in early adulthood. Adult endocrine and behavioural parameters were uniquely altered by manipulations in adolescence, but only behavioural effects were specific to cat odour priming. PPO enhanced adult responses to novelty but increased activity during cat odour stress testing. These findings indicate that adolescence is a sensitive period for stress response programming, and a potential mechanism for this could involve developmental alterations in mesocorticolimbic dopamine function.

T47) Effects of environmental manipulations on maternal behaviour and implications for the development of the rodent limbic system during the first week of life.

Rahia Mashoodh, Stephanie Nelson, Chris J. Sinal, Tara S. Perrot-Sinal

Differences in expression of rat maternal behaviours are associated with differential development of endocrine, emotional, and cognitive responses to stress in offspring. It is hypothesized that plasticity of maternal behaviour serves as the interface between offspring development and the current environment demands. Consequently, offspring neural and behavioural profiles will be optimally programmed to suit the environment in which they are reared. In the first condition dams were exposed to a predation threat (cat odour) on the day of birth. To further gauge the extent to which environmental complexity modulated mother-infant interactions, pregnant dams were introduced to an enriched environment on GD16. Dams and pups lived in this environment until PND7, when pups were sacrificed, brains removed and hippocampi dissected. Using quantitative PCR, changes in excitatory GABA and expression of the transcription factor NGFI-A were used to measure the developmental consequences of these maternal environments.

T48) Four sources of error in the neuro-behavioural analysis of inbred, mutant, and transgenic mice.

Richard Brown and Tim O'Leary

When we began to develop behavioural test batteries for mice over 10 years ago, we were naïve and, like many others, we thought that you just bought mice and tested them, using the apparatus and procedures published in journal articles. But we soon found out that this is not as simple as it sounds. We have discovered four major sources of error in our experiments and I will discuss how we have dealt with these problems. They are:

- (1) The mouse problem. Mice have a mind of their own and do not want to do what you want them to do. Mouse strains differ in many unexpected ways. How do you determine the problems with each strain?
- (2) The apparatus problem. Many types of apparatus for mice do not do what people claim in their papers, and modification of the apparatus gives different results. How do you determine which version of an apparatus to use?
- (3) The procedure problem. There are many ways of using an apparatus, and different procedures give different results. Which procedure should be used?
- (4) The experimenter-error problem. Experimenters make mistakes. How do you detect them, correct them, and eliminate them?

POSTERS

P1) The Role of Superordinate Labels on Categorical Flexibility in 3- and 4-year-olds

Carly Prusky & Sophie Jacques

The purpose of the current study was to investigate the effect of superordinate labels on categorical flexibility in 3- and 4-year-olds using two match-to-sample tasks that assess categorization and cognitive flexibility. It was expected that both 3- and 4-year-olds would perform better on the categorization task than on the cognitive flexibility task. In addition, 4-year-olds were expected to improve in a label condition on both the categorization task and the cognitive flexibility task, but 3-year-olds were not. As predicted, 4-year-olds performed better than 3-year-olds on both versions of both tasks and performance was better on the categorization task than on the cognitive flexibility task for both age groups. Surprisingly, however, 4-year-olds did not benefit from labels on either task but 3-year-olds did. Results are discussed in terms of the role that different labels might have on categorical flexibility.

P2) Improvements in Quality of Life in Patients with Obstructive Sleep Apnea using Continuous Positive Airway Pressure (CPAP) Treatment is Related to Decreased Sleepiness and Improved Mood.

E.L. Laur¹, E.Y.Y. Lau^{1,2}, G.A. Eskes^{1,2}, D. Morrison^{1,2}, M. Rajda^{1,2}, and K. Spurr².

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Obstructive Sleep Apnea (OSA) is a disorder characterized by brief breathing disruptions during sleep. Symptoms include excessive daytime sleepiness, depression, and decreased memory, attention and quality of life (QoL). Continuous Positive Airway Pressure (CPAP) is the standard treatment for OSA. CPAP keeps the airway open to eliminate disrupted breathing and is effective in restoring sleep and improving daytime symptoms. The current study aimed to investigate the role of sleepiness and mood on QoL in patients with OSA. We hypothesized that both mood and residual sleepiness are associated with QoL after using CPAP. Thirty patients with moderate to severe OSA were asked to complete questionnaires to measure sleepiness, mood, and QoL. CPAP was found to significantly improve QoL ($t(24)=-8.977$, $p=0.00$) and decrease sleepiness ($t(28)=6.323$, $p=0.00$). QoL was negatively correlated with sleepiness $r(29)=-0.605$, $p=0.00$) and depressed mood $r(29)=-0.629$, $p=0.00$). Our findings suggest that the effects of CPAP treatment on sleepiness and depression are associated with improvement in QoL.

P3) Sleep Problems and Performance on a New Test of Attention in Children with and without ADHD

Jennifer C. Mullane, MA, Jenna L. Bower, BSc, & Penny V. Corkum, PhD

Few studies have examined the relationship between sleep disturbances and attention in children and even fewer have investigated this association in children with ADHD. This pilot study examined differences in sleep problems between children with and without ADHD and explored the effects of sleep disturbances on three networks of attention as measured by the Attention Network Test (ANT). Participants were 24 children with ADHD-Combined Type (Mage = 110.58 mths), 17 children with ADHD-Inattentive Type (Mage = 116.12 mths), and 45 typically developing children (Mage = 115.32 mths). Children with both subtypes of ADHD had significantly more sleep problems than controls, as measured by the Children's Sleep Habits Questionnaire. Partial correlation analyses found that sleep disturbances were related to less efficient performance on the ANT's measure of executive attention, but only in the ADHD-Combined Type group.

P4) Children's Attributions for Ambiguous Intentions: Hostile or Outcome Driven?

Ashley Brown & Sophie Jacques

The current study examined three possible reasons for why aggressive children make hostile attributions of intent in ambiguous situations: social learning deficits, theory of mind deficits, or innate predispositions. Thirty-five typically developing children between the ages of 44 and 87 months were assessed on their tendency to make hostile attributions using an extended version of the hostile attribution bias interview. This extended interview included vignettes with both positive and negative outcomes as well as hostile, ambiguous, and benign intentions. Results indicated a main effect of intent and a trend toward a significant age group effect, but no effect of story outcome and no significant interactions. Results are discussed in terms of existing theories and limitations of the current study.

P5) How Does the Reader Read –er. Using the Missing Letter Task to Investigate Lexical Access of Morphologically Complex Words

Hilary D. Duncan, S. H. Deacon, Jean Saint-Aubin

Dual-route models of complex word processing posit two access routes to the lexicon: a direct-access route, wherein the word's full-form representation is accessed via the whole word; and a decompositional route, wherein the component morphemes are accessed separately. Evidence for a dual-route model comes from lexical decision tasks and eye-movement monitoring during fluent reading. These studies have shown that the time needed to identify morphologically complex words is influenced by whole-word and component

frequencies. The aim of the current study is to investigate the ability of the Missing Letter Effect task to replicate these results. Sixty-three university students were asked to search for the target letter –r while reading a prose passage containing high and low frequency one-morpheme and derived words. More omissions were made on one-morpheme than derived words when they were low-frequency, but there was no difference when they were high-frequency. Results support recent eye-movement studies that posit a dual-route model of lexical access.

P6) Does the risk of memory loss influence decision making by physicians and patients who contemplate temporal lobe resection for control of seizures?

Jaime Williams^{1,2} Jordan Green³, and Jeannette McGlone^{1,3} Capitol District Health Authority, Halifax, NS¹, Department of Psychology, University of Regina², Department of Psychology, Dalhousie University³

Temporal lobe resection (TLR) is a successful, although underutilized treatment for complex partial seizures. One reason for this underutilized may be the perceived risk of severe and permanent memory loss. However, patients' and physicians' perceptions of the risk of memory loss following TLR have not been empirically investigated. In this presentation, we propose a series of studies using a triangulation of methodologies examining whether fear of memory loss contributes to decision-making about elective brain surgery. First, we propose conducting focus groups with non-surgical, pre and post TLR patients to uncover their thoughts and attitudes towards TLR surgery and the risk of memory loss. Second, we will survey epilepsy specialists querying their thoughts regarding memory loss including how they communicate the risk to patients and what role the risk has in recommending and proceeding with TLR. Finally, we will explore information available on the Internet regarding TLR and the risk of memory loss, including the accuracy and consistency of the information. Implications of the research will be discussed including the importance of effective epilepsy-related knowledge transfer and critical comments will be invited.

P7) Selective cholinergic basal forebrain lesions using anti-murine-p75-saporin (mu p75-saporin): a mouse model of Alzheimer's disease?

Nicola Hoffman, Kazue Semba, Jennifer Stamp, Richard Brown

Degeneration of cholinergic neurons in the basal forebrain (BF) is a prominent characteristic of Alzheimer's disease (AD). One current animal model developed to mimic BF cholinergic neuron atrophy observed in human AD is via injecting neurotoxins into the mouse brain. Our long-term research aims are to selectively lesion mouse BF cholinergic neurons and then examine the cognitive and other functional deficits associated with this cell loss. Mu p75-saporin is an immunotoxin that has been shown to selectively destroy cholinergic neurons in the medial septum (MS), but demonstrates concurrent cell loss in cerebellar Purkinje cells. In the present study we examined the efficacy of the recently developed, presumably improved mu p75-saporin in selectively lesioning BF cholinergic neurons in the mouse brain. One of three doses (1.0, 1.5, or 2.0 µg/µl) of the immunotoxin or vehicle was infused bilaterally into the lateral ventricles of male C57BL/6J mice. Two weeks following injections, mice were intra-cardially perfused, the brains removed and sectioned for immunohisto-

chemistry to visualize cholinergic and a subtype of GABAergic BF neurons, as well as acetylcholinesterase (AChE) histochemistry. Quantitative analysis revealed a reduction of cholinergic neurons in the MS/VDB following intracerebroventricular injections of mu p75-saporin, but only a trend for a decrease of cholinergic neurons in the nBM. The density of AChE-positive fibers in the hippocampus and somatosensory cortex was not significantly altered following neurotoxin injection, although there was a trend for a decrease in the hippocampus. No loss of noncholinergic neurons in the MS or nBM was apparent, nor was there a loss of noncholinergic cerebellar Purkinje cells. The results indicate that the most recently developed mu p75-saporin has an increased selectivity to cholinergic MS neurons, compared with the previous formula, and may be useful as a selective cholinergic immunotoxin for mice. Mu 75-saporin may be a useful tool to create an animal model of AD and expand research concerning the progression and neuropathology of the disease.

P8) “Don’t Look at a White Bear!” Action Suppression in Adults

Teri Bolivar & Sophie Jacques

The purpose of this study was to develop an objective measure of action suppression. In this task, participants were required to inhibit looking at the *content* of a picture cue, while still using its location to inform them about the *location* of target words to which they had to respond. Although reaction time data showed few effects, accuracy data showed that participants were unable to ignore the content of cues at SOA of 300 ms. Results are discussed in relation to prior thought suppression research and findings in the attention literature (Wegner et al., 1987; Wegner & Zanakos, 1994; Jonides, 1981; Danziger & Kingstone, 2005).